

Weather definitions with Hogan

Bob

28 July 2026

Listen first

Hogan frames the question

- How do you see the weather question now?
- What would make the later-period comparison useful?
- What should we simplify, correct, or protect?

Bob listens and writes

- Put genuine choices on the table
- Record Hogan's wording
- Read it back before leaving

Credit. Li et al. provide the *Atmospheric Research* heatwave event family. Hogan proposed the project adaptation: count events by month, then identify upper-tail months. Hogan leads the weather framing.

One named family; different estimands

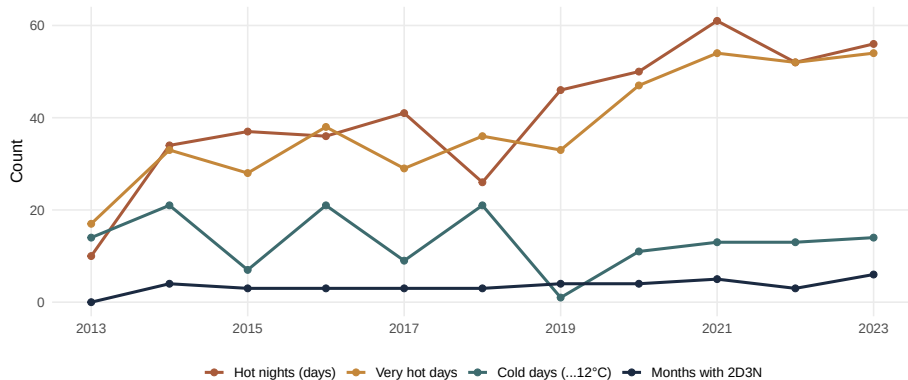
Jasmine confirmed	Liu J et al. (2020): Hong Kong daily mortality. Reported attributable fractions were 4.72% for cold and 0.16% for heat .
Roro's siblings	HWD_Tavg, HWD_Tmax, HWD_Tmin, and HWD_Tcombined. Keep all four as distinct named candidates.
Current project	Monthly thermal exposures paired with true-event-month stroke aggregates, subject to Roro's governed outcome rules.

The evidence is cumulative, not interchangeable. Tuesday chooses exposure definitions before any project health results are inspected.

Heat and cold coexist across the study window

Heat rose; cold did not vanish

HKO Headquarters extremes + Wang/Ren-style 2D3N months, 2013...2023



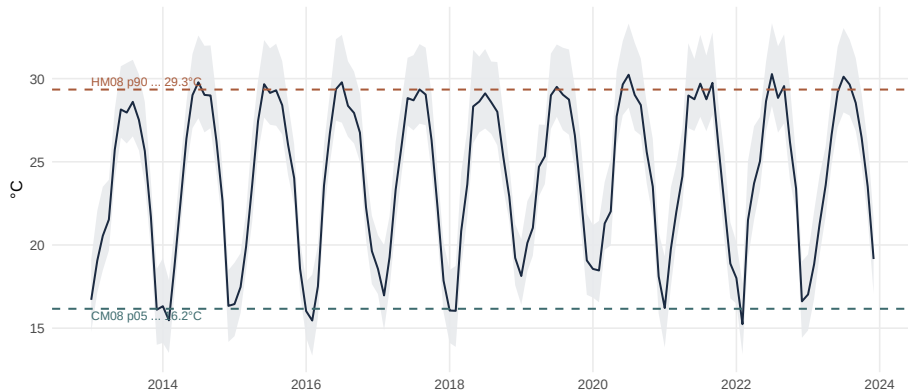
Environmental descriptives only. Weather definitions to be locked with Hogan.

Exposure-only description for discussion with Hogan; no stroke findings.

Relative tails depend on the reference freeze

Monthly mean temperature and relative tails

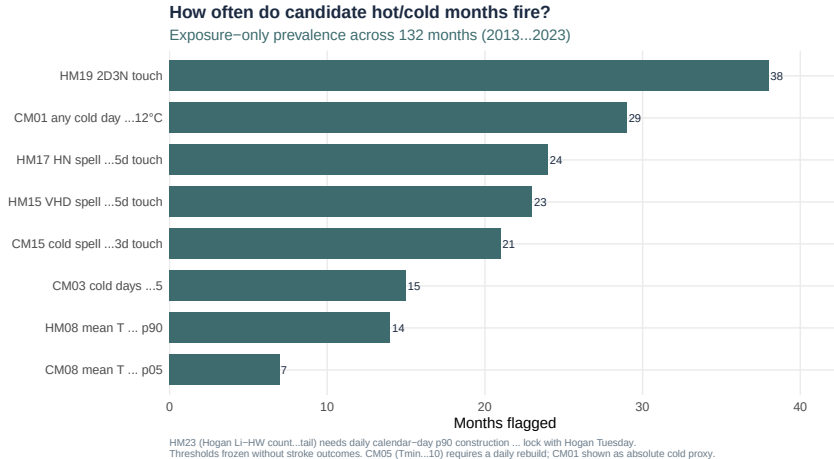
Ribbon = mean Tmin...Tmax; dashed lines = study-period monthly p90 / p05



Reference-period freeze is a Tuesday decision with Hogan (2013...2019 vs full window).

Decision needed: 2013–2019, full 2013–2023, or a defensible external climatology?

Prevalence is a design check, not a winner



Counts help test usability. They do not choose a definition by themselves.

Starter hot-month and cold-month set

Hot-month starters		Cold-month starters	
HM23	Li-HW event-count upper tail; lock with Hogan	CM03	At least five days with $T_{\min} \leq 12^{\circ}\text{C}$
HM08	Monthly mean temperature at or above frozen p90	CM08	Monthly mean temperature at or below frozen p05
HM15	Touches at least five consecutive $T_{\max} \geq 33^{\circ}\text{C}$ days	CM15	Touches at least three consecutive $T_{\min} \leq 12^{\circ}\text{C}$ days
HM17	Touches at least five consecutive $T_{\min} \geq 28^{\circ}\text{C}$ nights		
HM19	Touches a Wang/Ren-style 2D3N window		

First-wave sensitivities: HM27 nighttime heat-degree burden; HM32 heat + ozone; CM05 any $T_{\min} \leq 10^{\circ}\text{C}$ day; CM30 cold + influenza. CM05 stays a labelled severe-cold sensitivity, not biology.

Six decisions to lock together

☐ **Li-HW / HM23**

Season, p90 operator, duration, gaps, monthly assignment, and tie rule.

☐ **Gate 3 pair**

Name one hot and one cold co-primary.

☐ **Roro siblings**

Place all four HWD_* definitions in the named family?

☐ **Reference freeze**

Choose the primary threshold period and percentile method.

☐ **CM05 role**

Keep it as severe-cold sensitivity only?

☐ **Age visibility**

Retain 65–69 and 70–74 if governance permits?

Write each decision in Hogan's words. If something stays open, record the precise question, owner, and date.

Thank you, Hogan

For leading the weather side and helping make it clearer.

- Read back the six written decisions
- Correct the wording together
- Send the short record to Hogan and Roro

No rush on anything that needs a little more thought :)